ALL INDIA TEST SERIES

FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FULL TEST – I

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A projectile is fired with velocity v at angle of θ with horizontal. What would be the radius of curvature of path at instant when rate of change of speed of projectile is minimum.

(A) $\frac{v^2\cos^2\theta}{g}$ (B) $\frac{v^2}{g\cos\theta}$

(C) $\frac{v^2\cos\theta}{g}$ (D) $\frac{v^2}{g\cos^2\theta}$

2. Position of particle is given by $x = t^3 - 4t^2 + 5t + 9$. What would be the distance travelled by particle from instant t = 0 to instant when particle changes its direction of velocity for last time.

(A) 1.85 m (B) 2.15 m

(C) 2m (D) 2.85 m

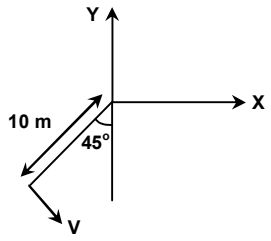
3. If a particle have velocity and acceleration given $\left(3\hat{i}+4\hat{j}\right)ms^{-1}, \left(4\hat{i}+3\hat{j}\right)ms^{-2}$ respectively.

What would be the rate of change of speed at that instant?

(A) 4.8 ms^{-2} (B) 5 ms^{-2}

(C) 3 ms^{-2} (D) 4 ms^{-2}

4. A ball of mass 1 kg attached with string of length 10 m is free to rotate in horizontal plane is subjected to constant force of $-\sqrt{2}\left(\hat{i}+\hat{j}\right)$N. What should be the velocity of projection at position shown in figure. Such that it complete full revolution.

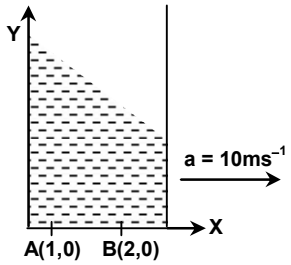


(A) 10 ms^{-1} (B) $2\sqrt{5}$ ms^{-1}

(C) $4\sqrt{5}$ ms^{-1} (D) 6 ms^{-1}

Space for Rough work

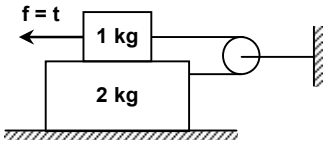
5. Fluid of density 10^3 kg/m^3 is kept in container. Container is moving with acceleration of 10 ms^{-2} find the magnitude of pressure difference between A and B as shown in figure.



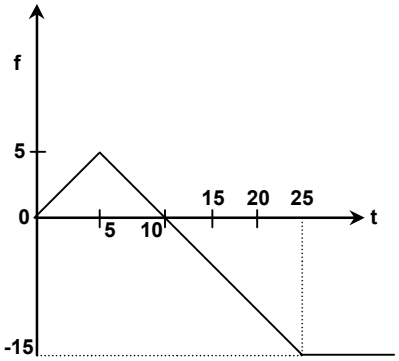
(A) 10^4 (B) 0

(C) $\sqrt{2}\times10^4$ (D) 2×10^4

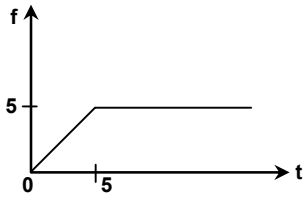
6. Coefficient of friction between all the surfaces is 0.5. Which diagram would describe the friction between ground and 2 kg block when f = t. N is applied on 1 kg block. It is given that string is not taut initially



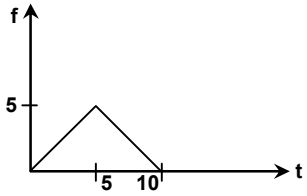
(A)



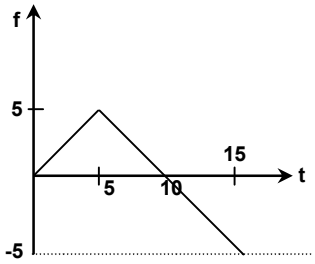
(B)



(C)

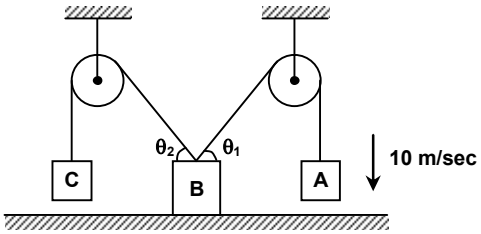


(D)



Space for Rough work

7. If v_A = 10 m/sec what would be the velocity of C. $\theta_1 = 30^0$, $\theta_2 = 60^o$. If B is very heavy and remains on ground.



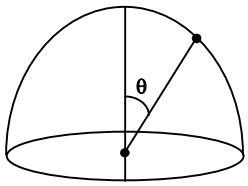
(A) $\frac{10}{\sqrt{3}}$ m/sec

(B) $10\sqrt{3}\ ms^{-1}$

(C) $10\ ms^{-1}$

(D) $5\sqrt{3}\ ms^{-1}$

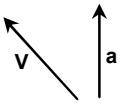
8. A ball of mass m kept at top slide freely on frictionless hemispherical surface. At what minimum angle θ net force acting on ball would be equal to mg

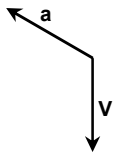


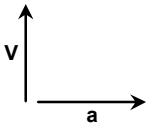
(A) $\cos^{-1}(2/3)$ (B) $\cos^{-1}(1/2)$

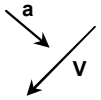
(C) 0° (D) 30°

9. From the given velocity acceleration diagram identify in which case object is slowing and turning to the left.

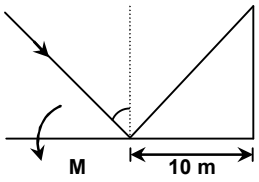
(A) 

(B) 

(C) 

(D) 

10. A light ray is incident on a plane mirror M. The mirror is rotated in direction as shown in figure by an arrows at frequency $(9/\pi)$ revolution/sec. The light reflected by the mirror is received on the wall W at a distance of 10 m from axis of rotation. speed of the spot on wall when angle of incident = 37° is



(A) 10 m/sec (B) 1000 m/sec

(C) 360 m/sec (D) 500 m/sec

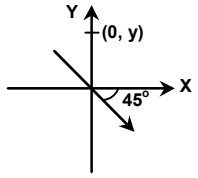
Space for Rough work

11. A ray of light moving along the vector $\left(\hat{i}-2\hat{j}\right)$ undergoes refraction at an interference of two media which is x-z plane μ for y > 0 is 2 while Y < 0 it is $\frac{\sqrt{5}}{2}$. The unit vector along which the refracted ray moves is

(A) $\frac{\left(-3\hat{i}-5\hat{j}\right)}{\sqrt{34}}$ (B) $\frac{-4\hat{i}-3\hat{j}}{5}$

(C) $\left(-\frac{3\hat{i}-4\hat{j}}{5}\right)$ (D) $\left(\frac{4\hat{i}-3\hat{j}}{5}\right)$

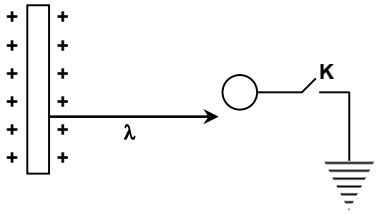
12. A dipole is placed at origin of co-ordinate system as shown in figure. Electric field at point P (0, y) is given as



(A) $\frac{K\rho}{\sqrt{3}y3}\left(-\hat{i}-\hat{j}\right)$ (B) $\frac{K\rho}{y^3}\left(-\hat{i}+\sqrt{2}\hat{j}\right)$

(C) $\frac{K\rho}{y^3}\left(-\hat{i}-2\hat{j}\right)$ (D) $\frac{K\rho}{\sqrt{2}y^3}\left(-\hat{i}-2\hat{j}\right)$

13. Figure shows a large conducting sheet having charge density σ on both sheet. The surface potential of sheet is V_0. A sphere of radius r is placed at a distance λ (λ >> r) from the sheet. When the key is closed charge q appears on the surface. The values σ on sheet is given by

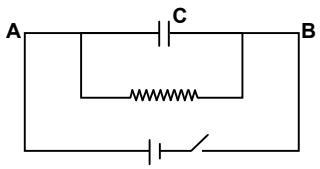


(A) $\left(\frac{q+4\pi\varepsilon_0 rV_0}{4\pi r\ell}\right)$ (B) $\left(\frac{4\pi\varepsilon_0 rV_0+q}{2\pi r\ell}\right)$

(C) $\left(\frac{4\pi\varepsilon_0 rV_0-q}{2\pi r\ell}\right)$ (D) Zero

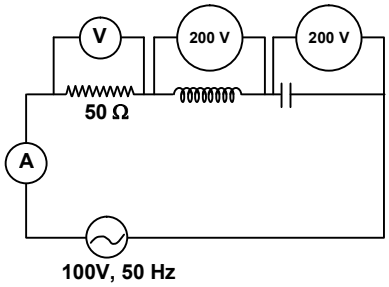
Space for Rough work

14. The resistance between A and B of given circuit at t = 0 and t = ∞ will be (it the switch is closed at t = 0).



(A) ∞ at both t
(B) O at both t
(C) O and R respectively
(D) R at both t

15. In series LCR circuit voltmeter and ammeter reading are



(A) V = 250 V, I = 4A
(B) V = 150 V, I = 2A
(C) V = 1000V, I = 5A
(D) V = 100 V, I = 2A

16. A radioactive substance X decays into another substance Y. Initially angle X was present. λ_x and λ_y are distingration constant of x and y. N_x and N_y are the number of nuclei of x and y at any time t. Number of nuclei N_y will be maximum when

(A) $\frac{N_y}{N_x - N_y} = \frac{\lambda_y}{\lambda_x - \lambda_y}$ (B) $\frac{N_x}{N_x - N_y} = \frac{\lambda_x}{\lambda_x - \lambda_y}$

(C) $\lambda_y N_y = N_x \lambda_x$ (D) $\lambda_x N_x = \lambda_x N_y$

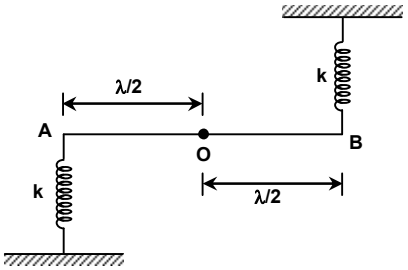
17. A capillary tube is immersed vertically in water and the height of the water column is x. When this arrangement is taken into a mine of depth d, the height of the water column is y. If R is the radius of the earth, the ratio $\frac{x}{y}$ is

(A) $\left(1-\frac{d}{R}\right)$ (B) $\left(1+\frac{d}{R}\right)$

(C) $\left(\frac{R-d}{R+d}\right)$ (D) $\left(\frac{R+d}{R-d}\right)$

Space for Rough work

18. The ends of a rod of length λ and mass m are attached to two identical springs as shown in figure. The rod is free to rotate about its centre O. The rod is depressed slightly at end A and released. The time period of the resulting oscillation is



(A) $2\pi\sqrt{\frac{m}{2k}}$ (B) $2\pi\sqrt{\frac{2m}{k}}$

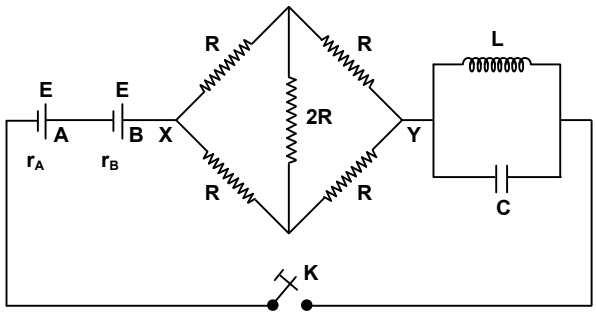
(C) $\pi\sqrt{\frac{2m}{3k}}$ (D) $\pi\sqrt{\frac{3m}{2k}}$

19. The period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle which moves without friction down an inclined plane of inclination α, is given by

(A) $2\pi\sqrt{\frac{L}{g\cos\alpha}}$ (B) $2\pi\sqrt{\frac{L}{g\sin\alpha}}$

(C) $2\pi\sqrt{\frac{L}{g}}$ (D) $2\pi\sqrt{\frac{L}{g\tan\alpha}}$

20. In the circuit shown in figure, A and B are two cells of the same emf E and of internal resistances r_A and r_B respectively. L is an ideal inductor and C is an ideal capacitor. The key K is closed. When the current in the circuit becomes steady, what should be the value of R so that the potential difference across the terminals of cell A is zero



(A) $R = r_A - r_B$ if $r_A > r_B$

(B) $R = \sqrt{r_A r_B}$

(C) $R = \frac{1}{2}(r_A + r_B)$

(D) For no value of R will the potential difference between the terminals of cell A be equal to zero

Space for Rough work

21. A uniform metal rod is moving with a uniform velocity v parallel to a long straight wire carrying a current I. The rod is perpendicular to the wire with its ends at distance r_1 and r_2 (with $r_2 > r_1$) from it. The emf induced in the rod is

(A) Zero

(B) $\frac{\mu_0 Iv}{2\pi}\log_e\left(\frac{r_2}{r_1}\right)$

(C) $\frac{\mu_0 Iv}{2\pi}\log_e\left(\frac{r_1}{r_2}\right)$

(D) $\frac{\mu_0 Iv}{4\pi}\left(1-\frac{r_1}{r_2}\right)$

22. When a centimeter thick surface is illuminated with light of wavelength λ, the stopping potential is V. When the same surface is illuminated by light of wavelength 2λ, the stopping potential is V/3. The threshold wavelength for the surface is

(A) $\frac{4\lambda}{3}$

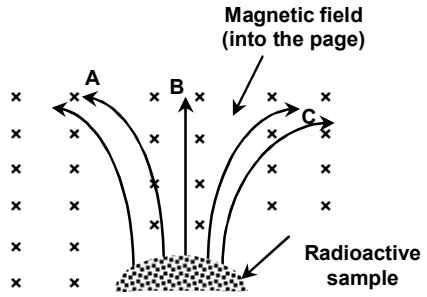
(B) 4λ

(C) 6λ

(D) $\frac{8\lambda}{3}$

23. The radiations emitted from a radioactive material separate into three distinct groups A, B and C when a magnetic field is directed into the plane of the paper. The names of radiations A, B and C, respectively, are (see figure)

(A) α, β and γ
(B) α, γ and β
(C) β, α and γ
(D) γ, α and β



24. The radioactivity of a sample is X at a time t_1 and Y at a time t_2. If the mean life of the specimen is τ, the number of atoms that have disintegrated in the time interval $(t_2 - t_1)$ is

(A) $X t_1 - Y t_2$
(B) $X - Y$
(C) $(X - Y)/\tau$
(D) $(X - Y)\tau$

25. A block of silver of mass 4 kg hanging from a string is immersed in a liquid of relative density 0.72. If relative density of silver is 10, then tension in the string will be: [Take g = 10 m/s^2]

(A) 37.12 N
(B) 42 N
(C) 73 N
(D) 21 N

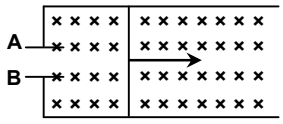
Space for Rough work

26. An electron is accelerated from rest through potential difference V. Then electron experience force f in uniform magnetic field on increasing the p.d. to v′ force became 2f. (V'/V) is equal to
(A) 1/4 (B) 2
(C) 1/2 (D) 4

27. Magnetic flux linked with a stationary loop of resistance R varies with time T as ϕ = at (T –t) amount of heat generated in loop during that time is:
(A) $\frac{aT}{3R}$ (B) $\frac{a^2T^2}{3R}$
(C) $\frac{a^2T^2}{R}$ (D) $\frac{a^2T^3}{3R}$

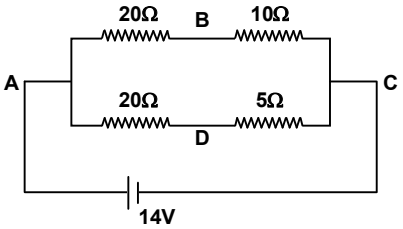
28. A large tank filled with water to height h is to be emptied through a hole at bottom ratio of time taken to value of water level from $\left(n \text{ to } \frac{n}{2}\right)$ & $\left(\frac{n}{2} \text{ to } 0\right)$ is
(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$
(C) $\sqrt{2}-1$ (D) $\frac{1}{\sqrt{2}-1}$

29. A conductivity rod of length = 1.0 m is moving with uniform speed v = 20m/sec in field of 4.0T. C = 10 μt is connected as shown in figure



(A) q_A = +80 μC, q_B = –80 μC
(B) q_A = –80 μC, q_B = +80 μC
(C) q_A = –40 μC, q_B = + 40 μC
(D) q_A = +40 μC, q_B = –40 μC

30. What resistor should be connected in parallel with 20Ω resistor in ADC branch so that potential difference between B & D may be zero.
(A) 20 Ω
(B) 10Ω
(C) 5Ω
(D) 15Ω



Space for Rough work

Chemistry

PART – II

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. The Vander Waal's equation of corresponding states for 1 mole of gas is:
[Where P_c = critical pressure, T_c = critical temperature, V_c = critical volume]

$\left[\pi = \frac{P}{P_c},\ \phi = \frac{V}{V_c} \text{ and } \theta = \frac{T}{T_c}\right]$

(A) $\left[\pi - \frac{3}{\phi}\right](3\phi - 1) = 8R\theta$

(B) $\left[\pi + \frac{3}{\phi}\right](3\phi - 1) = 8R\theta$

(C) $\left[\pi + \frac{3}{\phi}\right](3\phi + 1) = 8R\theta$

(D) $\left[\pi - \frac{\phi}{3}\right](3\phi - 1) = 8R\theta$

2. Given that $E^0_{Fe^{+2}|Fe} = -0.44V;\ E^0_{Fe^{+3}|Fe^{+2}} = 0.77$. If Fe^{+2}, Fe^{+3} and Fe solid are kept together then

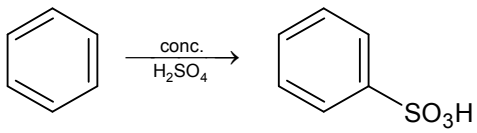
(A) $\left[Fe^{+3}\right]$ increase

(B) $\left[Fe^{+3}\right]$ decrease

(C) Fe^{+2}/Fe^{+3} remain unchanged

(D) $\left[Fe^{+2}\right]$ decrease

Space for Rough work

3 

Mark out the incorrect statement:
(A) Hexa deutarated benzene gets sulphonated slower than simple benzene
(B) Electrophile in the reaction is SO_3
(C) Dilution of acid during reaction promotes reversal (desulphonating)
(D) Polysulphonation is hard to proceed

4. To maintain the pH of 7.4 for blood at normal condition which is 2M in H_2CO_3 (at equilibrium). What volume of 5M $NaHCO_3$ solution is required to mix with 10ml of blood? $K(H_2CO_3) = 7.8 \times 10^{-7}$
(A) 78.36 ml (B) 102 ml
(C) 52.71 ml (D) 89.01 ml

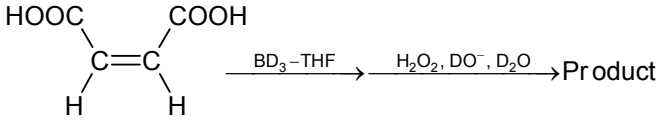
5. For fairly concentrated solution of a weak electrolyte A_xB_y, having concentration 'C' the degree of dissociation is given by:
(A) $\alpha = \sqrt{k_{eq} xy / c}$ (B) $\alpha = \sqrt{K_{eq} c / xy}$
(C) $\alpha = \sqrt{K_{eq} / c(x+y)}$ (D) $\alpha = K_{eq} / \left[c^{x+y-1}.x^x.y^y \right]^{1/x+y}$

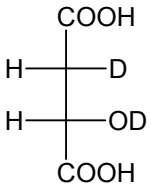
6. XeF_6 dissolves in anhydrous HF to give a good conductivity solution which contains:
(A) $H^{\oplus}$ and XeF_7^- ion (B) HF_2^- and $XeF_5^{\oplus}$
(C) $HXeF_6^{\oplus}$ and F^- ion (D) None

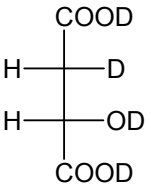
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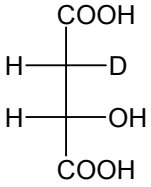
7. When the concentration of A is 0.1 M, it decomposes to give 'x' by a first order process with a rate constant of $6.93 \times 10^{-2} min^{-1}$. The reactant 'A' in the presence of catalyst gives 'Y' by a second order mechanism with a rate constant of 0.2 $min^{-1}M^{-1}$. In order that half life of both the process is 10 min. One should start the initial concentration of A as:
(A) 0.01M (B) 5.0 M
(C) 10.0 M (D) 0.5 M

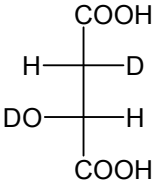
8. According to Arrhenius equation, the rate constant (K) and energy of activation (E) of a reaction are related by:
(A) $A = Ke^{-E/RT}$ (B) $K = Ae^{E/RT}$
(C) $K = Ae^{-E/RT}$ (D) $K = -Ae^{E/RT}$

9. 

(A) 

(B) 

(C) 

(D) 

10. In alkaline medium, H_2O_2 acts as an oxidising agent in its reaction with:
(A) $Cr_2(SO_4)_3$ (B) Ag_2O
(C) $K_3[Fe(CN)_6]$ (D) $K_2Cr_2O_7$

Space for Rough work

11. (a) KNO_2 (p) $pH = \frac{1}{2}(pka + pka_2)$

(b) $\overset{\oplus}{Ph}NH_2Cl^-$ (q) $pH = \frac{1}{2}(pkw + pka - pkb)$

(c) $NaHCO_3$ (r) $pH = \frac{1}{2}(pkw - pkb - \log c)$

(d) NH_4CN (s) $pH = \frac{1}{2}(pkw + pka + \log c)$

For abcd correct order of matching is

(A) pqrs (B) qprs

(C) rspq (D) srpq

12. In which of the following case, increase in concentration of ion cause increase in E_{cell}?

(A) $Pt(H_2) | H^{\oplus}(aq)$ (B) $Ag, AgCl | Cl^-(aq)$

(C) $Pt | quinhydron | H^-$ (D) $Ag | Ag^{\oplus}(aq)$

13. Mark out the correct order of nucleophilicity in DMF:

(A) $\overset{\ominus}{CH_3} > \overset{\ominus}{OH} > \overset{\ominus}{NH_2} > F^{\ominus} > I^{\ominus}$ (B) $\overset{\ominus}{CH_3} > \overset{\ominus}{OH} > \overset{\ominus}{NH_2} > I^{\ominus} > F^{\ominus}$

(C) $I^{\ominus} > F^{\ominus} > \overset{\ominus}{OH} > \overset{\ominus}{NH_2} > CH_3^{\ominus}$ (D) $\overset{\ominus}{CH_3} > \overset{\ominus}{NH_2} > \overset{\ominus}{OH} > F^{\ominus} > I^{\ominus}$

14. Mark out the right combination of cell and conditious for spontenety

(A) $Pt\underset{P_1}{(H_2)} | \underset{1M}{HCl} | Pt\underset{P_2}{(H_2)};\ P_1 > P_2$

(B) $Zn | Zn^{2+}(C_1) || Zn^{+2}(C_1) | Zn;\ C_1 > C_2$

(C) $Pt(Cl_2,\ 1\,atm) | Cl^-(C_1) || Cl^-(C_2) | Pt(Cl_2, 1\,atm);\ C_2 > C_1$

(D) $Pt(H_2,\ 1\,atm) | HCl(C_1) || HCl(C_2) | Pt(H_2), 1\,atm;\ C_1 > C_2$

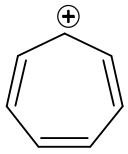
15. The dissolution of $Al(OH)_3$ by a solution of NaOH results in the formation of

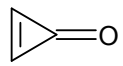
(A) $[Al(H_2O)_4(OH)_4]^+$ (B) $[Al(H_2O)_3(OH)_3]$

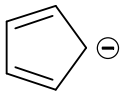
(C) $[Al(H_2O)_2(OH)_4]^-$ (D) $[Al(H_2O)_6(OH)_3]$

Space for Rough work

16. Which of the following is polar and aromatic?

(A) 

(B) 

(C) 

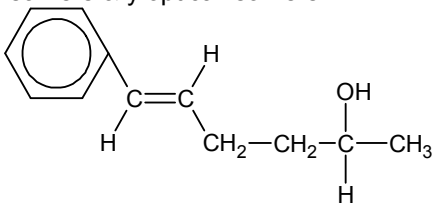
(D) None

17. The species that undergo disproportion in alkaline medium:

(A) Cl_2 (B) MnO_4^-

(C) HNO_3 (D) ClO_4^-

18. The compound, whose stereo chemical formula is written below, exhibits are geometrical isomers & y optical isomers



The value of x & Y are:

(A) 4 & 4 (B) 2 & 2

(C) 2 & 4 (D) 4 & 2

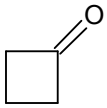
19. $Z-CH=CH_2 \xrightarrow{HBr} Z-CH_2-CH_2-Br$

Which of the following is Z?

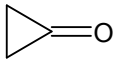
(A) —Cl (B) $—SO_3H$

(C) $—OCH_3$ (D) $—CH_3$

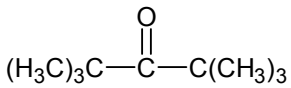
20. Rank the following in the order of increasing value of the equilibrium constant for hydration:



(I)



(II)



(III)

(A) I < II < III (B) III < I < II

(C) II < I < III (D) II < III < I

Space for Rough work

21. To a one liter solution of 0.1 N HCl, 0.025 mol of NH_4Cl are added. Assuming 80% dissociation of the solutes, the freezing point of the solution is (K_f = 1.85 deg/molal)
(A) –0.33°C (B) –0.85°C
(C) –0.23°C (D) –0.416°C

22. The suggested mechanism for reaction:

$$CHCl_3 + Cl_2 \longrightarrow CCl_4 + HCl$$

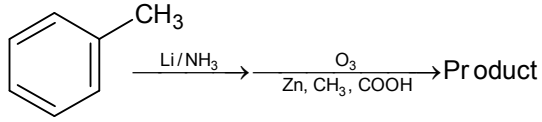
$$Cl_2 \underset{K_2}{\overset{K_1}{\rightleftharpoons}} 2Cl^{\bullet}$$

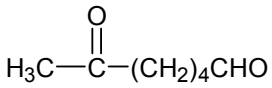
$$CHCl_3 + Cl^{\bullet} \xrightarrow{K_3} HCl + \dot{C}Cl_3 \,(\text{slow})$$

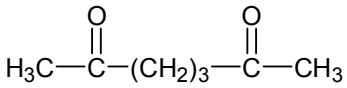
$$\dot{C}Cl_3 + \dot{C}l \xrightarrow{K_4} CCl_4$$

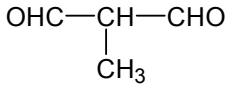
The experimental mechanism for reaction:
(A) $R = K_3[CHCl_3][Cl_2]$ (B) $R = K_3K_2[CHCl_3][Cl_2]^{1/2}$
(C) $R = K_3K_1[CHCl_3][Cl]$ (D) $R = K_3(K_{eq})^{1/2}[CHCl_3][Cl_2]^{1/2}$

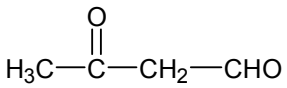
23. A catalyst increases the rate of reaction at 27°C by 10 times. By what amount it is decreasing E_a at the same temperature.
(A) 2 kJ/mol (B) 1.3 kJ/mol
(C) 5.7 kJ /mol (D) 21 kJ/mol

24. 

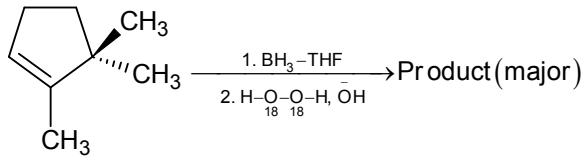
(A) 

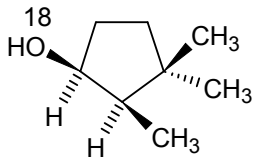
(B) 

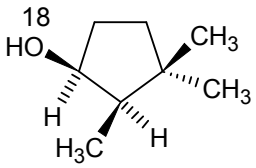
(C) 

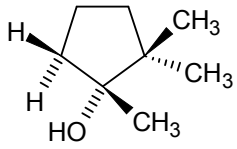
(D) 

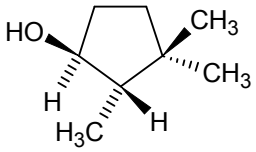
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25. 

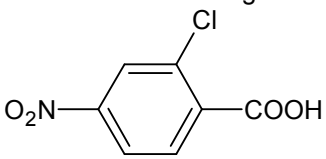
(A) 

(B) 

(C) 

(D) 

26. Which of the following is the best synthesis of 2-chloro-4-nitro benzoic acid?



(A) (i) Heat benzoic acid with $HNO_3 + H_2SO_4$
(i) Cl_2, $FeCl_3$, Δ

(B) (i) Toluene + ($HNO_3 + H_2SO_4$)
(ii) $K_2Cr_2O_7$, H_2O, H_2SO_4, Δ
(iii) Cl_2, $FeCl_3$

(C) (i) Heat chlorobenzene with ($HNCO_3 + H_2SO_4$)
(ii) CH_3Cl, $AlCl_3$
(ii) $K_2Cr_2O_7$, H_2O, H_2SO_4, Heat

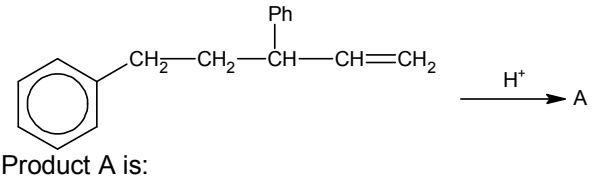
(D) (i) Toluene + ($HNO_3 + H_2SO_4$)
(ii) Cl_2, $FeCl_3$, Heat
(iii) $K_2Cr_2O_7$, H_2O, H_2SO_4, heat

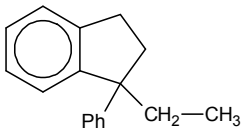
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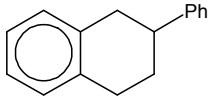
27. When a certain volume, V_1 ml of a gaseous hydrocarbon is exploded with excess of oxygen and the product coded to room temperature, the observed concentration in volume was 1.5 V_1 ml. A further contraction of $2V_1$ ml was observed on exposure of product to alkali. Formulae of hydrocarbon is
(A) C_2H_4 (B) C_2H_2
(C) C_2H_6 (D) C_3H_4

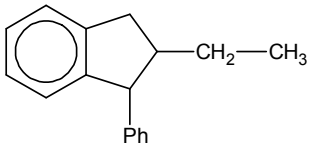
28. The de-broglie wavelength of an electron that has been accelerated through a potential difference of 100V is:
(A) 0.12264nm (B) 1.226m
(C) 1.226nm (D) 0.012264nm

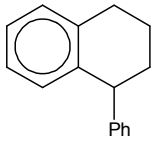
29. The correct order of increasing $C-O$ bond length of CO, CO_3^{2-}, CO_2 is:
(A) $CO_3^{2-} < CO_2 < CO$ (B) $CO_2 < CO_3^{2-} < CO$
(C) $CO < CO_3^{2-} < CO_2$ (D) $CO < CO_2 < CO_3^{2-}$

30. 

Product A is:
(A) 

(B) 

(C) 

(D) 

Space for Rough work

Mathematics

PART – III

SECTION – A
Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. For the system of equations

$\log_{10}\left(x^3 - x^2\right) = \log_5 y^2$

$\log_{10}\left(y^3 - y^2\right) = \log_5 z^2$

$\log_{10}\left(z^3 - z^2\right) = \log_5 x^2$

Which of following is/are true?
(A) there are infinite number of solution
(B) there is unique solution with x, y, $z \in Q$
(C) there are exactly two solution with x, y, $z \in Q$
(D) there is no solution

2. Three points A, B and C are considered on a parabola. The tangents to the parabola at these points form triangle MNP (NP being tangent at A, PM at B and MN at C). If the line through B and parallel to axis of parabola intersect AC at L then quadrilateral LMNP
(A) is always a parallelogram
(B) can never be parallelogram
(C) is parallelogram only when ordinates of A, B, C are in AP
(D) has exactly two sides parallel to each other, always

3. Let $f(x) = \dfrac{e^{\tan x} - e^x + \ln(\sec x + \tan x) - x}{\tan x - x}$ be a continuous function at x = 0. The value of f(0) equals

(A) $\dfrac{1}{2}$ (B) $\dfrac{2}{3}$

(C) $\dfrac{3}{2}$ (D) 2

Space for rough work

4. Let α, β, γ are the roots of the equation , $8x^3 + 1001x + 2008 = 0$. The value of $(\alpha + \beta)^3 + (\beta + \gamma)^3 + (\gamma + \alpha)^3$ is
(A) 251 (B) 751
(C) 735 (D) 753

5. Number of four digit numbers of the form N = abcd which satisfy following three condition
(i) $4000 < N < 6000$ (ii) N is a multiple of 5 (iii) $3 \le b < c \le 6$ is equal to
(A) 12 (B) 18
(C) 24 (D) 48

6. A coin that comes up head with probability $p > 0$ and tails with probability $1 - p > 0$ independently on each flip, is flipped eight times. Suppose the probability of three heads and five tails is equal to $\frac{1}{25}$ of the probability of five heads and three tails. The value of p is
(A) $\frac{5}{6}$ (B) $\frac{2}{3}$
(C) $\frac{1}{6}$ (D) $\frac{3}{4}$

7. The power of 'x' which has the greatest coefficient in the expansion of $\left(1+\frac{x}{2}\right)^{10}$ is
(A) 2 (B) 3
(C) 4 (D) 5

8. The first two terms of a G.P. add up to 12. The sum of the 3rd and 4th term is 48. If the term of the G.P. are alternatively positive and negative, then the first term is
(A) –4 (B) –12
(C) 12 (D) 14

9. If a curve is represented parametrically by the equations $x = 4t^3 + 3$ and $y = 4 + 3t^4$ and $\frac{\left(\frac{d^2x}{dy^2}\right)}{\left(\frac{dx}{dy}\right)^n}$ is a constant then the value of n, is
(A) 3 (B) 4
(C) 5 (D) 6

Space for rough work

10. The minimum value of the function $f(x) = x^{3/2} + x^{-3/2} - 4\left(x + \frac{1}{x}\right)$ for all permissible real x, is

(A) –10 (B) –6
(C) –7 (D) –8

11. The complete set of value of 'a' for which there exists at least one line that is tangent to the graph of the curve $y = x^3 - ax$ at one point and normal to the graph at another point is given by

(A) $a \in \left(-\infty, -\frac{4}{3}\right]$ (B) $a \in \left[-\frac{4}{3}, \infty\right)$
(C) $a \in \left[\frac{4}{3}, \infty\right)$ (D) $a \in \left(-\infty, \frac{4}{3}\right]$

12. In ΔABC, if $\frac{\sin A}{c \sin B} + \frac{\sin B}{c} + \frac{\sin C}{b} = \frac{c}{ab} + \frac{b}{ac} + \frac{a}{bc}$, then the value of A is (All symbols have there usual meaning)

(A) 120º (B) 90º
(C) 60º (D) 30º

13. If $A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal matrix, then |abc| is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{1}{6}$ (D) 1

14. If $A = \left[a_{ij}\right]_{n \times n}$, where $a_{ij} = i^{100} + j^{100}$, then $\lim\limits_{n \to \infty} \left(\frac{\sum\limits_{i=1}^{n} a_{ii}}{n^{101}} \right)$ equals

(A) $\frac{1}{50}$ (B) $\frac{1}{101}$
(C) $\frac{2}{101}$ (D) $\frac{3}{101}$

Space for rough work

15. If $a^2 + b^2 + c^2 + ab + bc + ca \le 0 \ \forall$ a, b, c $\in$ R, then value of the determinant

$$\begin{vmatrix} (a+b+2)^2 & a^2+b^2 & 1 \\ 1 & (b+c+2)^2 & b^2+c^2 \\ c^2+a^2 & 1 & (c+a+2)^2 \end{vmatrix}$$ is equal to

(A) 65 (B) $a^2 + b^2 + c^2 + 31$
(C) $4(a^2 + b^2 + c^2)$ (D) 0

16. If z is a complex number satisfying the equation $|z - (1 + i)|^2 = 2$ and $w = \frac{2}{z}$, then the locus traced by 'w' in the complex plane is
(A) x – y – 1 = 0 (B) x + y – 1 = 0
(C) x – y + 1 = 0 (D) x + y + 1 = 0

17. The angle between pair of tangents drawn to the curve $7x^2 - 12y^2 = 84$ from M(1, 2) is
(A) $2\tan^{-1}\frac{1}{2}$ (B) $2\tan^{-1} 2$
(C) $2\left(\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{2}\right)$ (D) $2\tan^{-1} 3$

18. Tangents PA and PB are drawn to the circle $x^2 + y^2 = 8$ from any arbitrary point p on the line
x + y = 4. The locus of the midpoint of chord of contact AB is
(A) $x^2 + y^2 + 2x + 2y = 0$ (B) $x^2 + y^2 - 2x - 2y = 0$
(C) $x^2 + y^2 - 2x + 2y = 0$ (D) $x^2 + y^2 + 2x - 2y = 0$

19. Let A(5, 12), B(–13 cos θ, 13 sin θ) and C(13 sin θ, –13 cos θ) are vertices of ΔABC where $\theta \in$ R. The locus of the orthocentre of ΔABC is
(A) x – y + 7 = 0 (B) x – y – 7 = 0
(C) x + y – 7 = 0 (D) x + y + 7 = 0

20. The range of the $f(x) = \sqrt{\cos^{-1}\left(\sqrt{\log_4 x}\right) - \frac{\pi}{2}} + \sin^{-1}\left(\frac{1+x^2}{4x}\right)$ is equal to

(A) $\left(0, \frac{\pi}{2} + \sqrt{\frac{\pi}{2}}\right]$ (B) $\left[\frac{\pi}{2}, \frac{\pi}{2} + \sqrt{\frac{\pi}{2}}\right]$
(C) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$ (D) $\left\{\frac{\pi}{6}\right\}$

Space for rough work

21. If $\lim\limits_{n\to\infty}\dfrac{n\cdot 3^n}{n(x-2)^n+n\cdot 3^{n+1}-3^n}=\dfrac{1}{3}$ where $n\in N$, then the number of integer(s) in the range 'x' is

(A) 3 (B) 4

(C) 5 (D) infinite

22. For any natural number m, $\int\left(x^{7m}+x^{2m}+x^m\right)\left(2x^{6m}+7x^m+14\right)^{1/m}dx$ (where x > 0), equals

(A) $\dfrac{\left(7x^{7m}+2x^{2m}+14x^m\right)^{\frac{m+1}{m}}}{14(m+1)}+c$ (B) $\dfrac{\left(2x^{7m}+14x^{2m}+7x^m\right)^{\frac{m+1}{m}}}{14(m+1)}+c$

(C) $\dfrac{\left(2x^{7m}+7x^{2m}+14x^m\right)^{\frac{m+1}{m}}}{14(m+1)}+c$ (D) $\dfrac{\left(7x^{7m}+2x^{2m}+x^m\right)^{\frac{m+1}{m}}}{14(m+1)}+c$

23. Let f: $(0,\infty)\to R$ be a continuous function such that $F(x)=\int_0^x tf(t)dt$. If $F(x^2)=x^4+x^5$, then $\sum\limits_{r=1}^{12}f\left(r^2\right)$ is equal to

(A) 216 (B) 219

(C) 222 (D) 225

24. The area of the region bounded by the curve $C: y=\dfrac{x+1}{x^2+1}$ and the line L : y = 1, is

(A) $1-\dfrac{1}{2}\ln 2+\dfrac{\pi}{4}$ (B) $\ln 2-\dfrac{\pi}{4}+1$

(C) $\dfrac{1}{2}\ln 2+\dfrac{\pi}{4}-1$ (D) $\ln 2-\dfrac{\pi}{2}+1$

25. If $\dfrac{dy}{dx}=\left(e^y-x\right)^{-1}$ where y(0) = 0, then y is expressed explicitly as

(A) $\dfrac{1}{2}\ln\left(1+x^2\right)$ (B) $\ln(1+x^2)$

(C) $\ln\left(x+\sqrt{1+x^2}\right)$ (D) $\ln\left(x+\sqrt{1-x^2}\right)$

Space for rough work

26. If $\vec{a}, \vec{b}, \vec{c}$ are non-zero vectors then value of the scalar $\left(\left(\vec{a}\times\vec{b}\right)\times\vec{a}\right)\left(\left(\vec{b}\times\vec{a}\right)\times\vec{b}\right)$ equals

(A) $-\left(\vec{a}\cdot\vec{b}\right)\left|\vec{a}\times\vec{b}\right|^2$ (B) $\vec{a}^2\left|\vec{a}\times\vec{b}\right|^2$

(C) $\left|\vec{b}\right|^2\left|\vec{a}\times\vec{b}\right|^2$ (D) $\left(\vec{a}\cdot\vec{b}\right)\left|\vec{a}\times\vec{b}\right|^2$

27. The distance between the line x = 2 + t, y = 1 + t, $z=-\frac{1}{2}-\frac{t}{2}$ and the plane $\vec{r}\left(\hat{i}+2\hat{j}+6\hat{k}\right)=10$, is

(A) $\frac{1}{6}$ (B) $\frac{1}{\sqrt{41}}$

(C) $\frac{1}{7}$ (D) $\frac{9}{\sqrt{41}}$

28. Let a relation R in the set N of natural numbers be defined as (x, y) $\in$ R if and only if $x^2 - 4xy + 3y^2 = 0\ \forall\ x, y \in N$. The relation R is

(A) reflexive (B) symmetric

(C) transitive (D) an equivalence relation

29. The top of a hill observed from the top and bottom of a building of height h is at angles of θ and ϕ respectively. The height of the hill is

(A) $\frac{h\cot\phi}{\cot\phi-\cot\theta}$ (B) $\frac{h\cot\theta}{\cot\theta-\cot\phi}$

(C) $\frac{h\tan\theta}{\tan\theta-\tan\phi}$ (D) none of these

30. Mean deviation about mean from the following data

x_i :	3	9	17	23	27
f_i :	8	10	12	9	5

is

(A) 7.15 (B) 7.09

(C) 8.05 (D) none of these

Space for rough work